Coding & Solution

Date	15 March 2026
Team ID	xxxxxx
Project Name	Credit Card Approval Prediction Using Machine Learning
Maximum Marks	5 Marks

Solution Summary

Field	Details
Repository Link / URL	https://github.com/Arsheenkhanam/Credit-Card-Approval-Prediction
Programming Language(s)	Python, HTML5, CSS3, JavaScript
Framework(s) Used	Flask, Bootstrap, Scikit-learn
Key Features Implemented	• Credit card approval prediction using Machine Learning. • Data preprocessing and feature encoding. • User-friendly web interface for applicant data entry. • Real-time prediction (Approved/Rejected). • Admin dashboard. • Model evaluation using Accuracy, Confusion Matrix, and Classification Report. • Responsive UI and input validation.
Pending / Incomplete Features	• User authentication for multiple user roles. • Prediction history storage in a database. • Email/SMS notification after prediction. • Explainable AI (reason for approval/rejection).
Setup / Run Instructions	1. Clone the GitHub repository. 2. Install required libraries using **pip install -r requirements.txt**. 3. Run the application using **python app.py**. 4. Open the provided local URL **(e.g., http://127.0.0.1:5000)** in a web browser. 5. Enter applicant details and click **Predict** to view the result.

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Code Quality Checklist

S.N o	Criteria	Status (Yes / No)
1	Code is modular and organized into functions / classes	Yes
2	Meaningful variable and function names are used	Yes
3	Code includes comments / documentation where necessary	Yes
4	Error handling is implemented for critical operations	Yes
5	The application runs without critical errors	Yes
6	Code is committed to a version control repository	Yes

Additional Notes / Comments:

The project follows a modular architecture using Flask for the backend and Scikit-learn for machine learning. Data preprocessing, model training, prediction, and the user interface are organized into separate components to improve readability, maintainability, and scalability. Version control is managed through GitHub, and the application has been tested to ensure accurate predictions and a smooth user experience.